Predicting the Sign of Asset Allocation Returns under Non-Stationarity

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Abstract. Our final model reaches 51.22% public accuracy on the QRT Data Challenge 167, against 50.79% for the organiser's LightGBM benchmark, and beats it by 0.56 points on a held-out period that was never used for any decision. The main contribution is methodological. The date labels of the dataset are shuffled; we reconstruct the true chronology from overlapping return windows and show that the gap between the benchmark's cross-validation (52.2%) and its leaderboard score comes from a signal that weakens over time, not from leakage. We therefore validate on the recent regime only, with purged date blocks, a pre-registered decision rule and a lockbox. The extra accuracy comes from cross-sectional information (what similar allocations and groups did yesterday), not from model complexity: linear, boosted and neural models hit the same ceiling. Across ten submissions, local validation ranks models the way the public leaderboard does (Spearman 0.82).

1. Problem and data

Each row is a pair (date t, allocation S), where an allocation is a long/short portfolio rebalanced daily with weights summing to 1 in absolute value. The task is to predict whether its return on day t+1 is positive. The metric is plain accuracy on the sign, so a constant "up" forecast already scores the base rate.

	Train	**Test**
Rows	527,073	31,870
Dates	2,522	120
Allocations	278	the same 278
Share of positive targets	50.7%	unknown

Features per row: the last 20 daily returns (`RET_1` most recent), the last 20 signed volumes, the median daily turnover and an anonymous group (1 to 4). The training target is the continuous return, which we use only through its sign. The test part used for the public leaderboard is roughly half of the test set, so a single public score carries about ±0.5 point of noise.

The signal is weak. The return of the previous day has a correlation of 0.085 with the target, and the rule "tomorrow has the sign of yesterday" scores 51.9% on the training set without any fitted parameter.

2. Shuffled dates and the reconstructed chronology

The labels `DATE_0001` to `DATE_2642` are not in calendar order. Volatility clusters in time in any real market, yet the lag-1 autocorrelation of the daily cross-sectional volatility along the label order is 0.03, the same as for a random permutation. A split on "the last 20% of dates" therefore has no temporal meaning.

The order can be recovered. The 20-day window of day b overlaps the window of the day before it, so `RET_2` of day b should match `RET_1` of day a across all allocations when b follows a. For every pair of dates we compute the average cross-sectional correlation between lag k of a and lag k+1 of b, for k = 1 to 10, normalised by the number of allocations present on both days. A link a → b is kept only when it is mutual (b is the best successor of a and a the best predecessor of b) and scores above 0.5.

The result is close to a full chronology of the training set: 4 chains of 1,780, 348, 256 and 130 consecutive dates, plus 8 isolated dates. Two checks that do not use the matching itself confirm it:

- `SIGNED_VOLUME_1` is filled on only one date in five. Along the chains, the filled dates recur every 5 positions in 92% of cases, the other gaps (4, 8, 9) matching a missing day such as a holiday. This flag is a day-of-week indicator.
- The number of allocations grows along the main chain, from 65 to 276, which fixes the direction of time.

The test set behaves differently: none of its 120 dates has a neighbour, either in the test set or in the training set. The organisers spaced the test dates so that the overlap cannot be used to read the test target. On the training set, the overlap gives the sign of the target in 88% of cases, which is why it matters so much for validation.

3. Validation protocol

The protocol was fixed before any model was fitted, and every result in this note uses the same folds.

1. **Lockbox.** The last 250 dates of the main chain, the most recent period we are sure of, are held out. The 20 dates before them are dropped so that no window overlaps the lockbox. The lockbox was opened once, for the final comparison in section 6.
2. **Purged date blocks.** The remaining dates are cut into blocks of 60 consecutive dates and spread over 8 folds, separately for old and recent blocks so that every fold contains recent dates. For each fold, training dates within 20 positions of a validation date of the same chain are removed: 20 is the window length, so no training row contains a return that is a validation target.
3. **Recent-regime metric.** Accuracy is measured on validation dates whose universe reaches 270 allocations (rolling median over 21 dates). This is the regime that resembles the test set, which has 278 allocations per date (section 4 explains why this matters).
4. **Decision rule.** An idea is kept only if it beats the current reference on the same 8 folds by more than two standard errors of the fold-by-fold difference. In the last rounds, after about 30 trials, the bar was raised to 2.5 standard errors and at least 7 folds out of 8.

All trials, including failures, were scored on the same folds (about 50 in total); the models kept or submitted are logged with their fold scores in a journal.

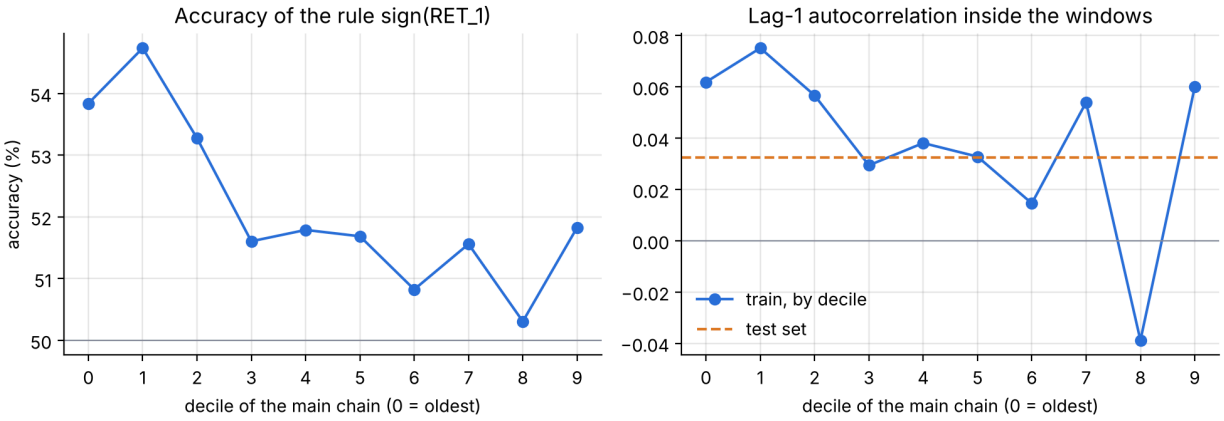
The protocol's first test was the organiser's benchmark itself. Scored with its original shuffled K-fold, with our blocks, and with our purged blocks, it gets the same accuracy on all dates (52.17% to 52.25%): purging changes nothing for row-wise features. What changes the score is the period being scored: 51.84% on recent dates, 51.42% on the lockbox, 50.79% on the public leaderboard. **The benchmark's optimism comes from non-stationarity, not from leakage.**

4. Where the signal is, and how it decays

Almost all the information sits in the previous day. The correlation of the target with `RET_k` is 0.085 for k = 1, slightly negative for k = 2 to 4 (down to -0.024) and close to zero beyond. The effect is mostly idiosyncratic: it keeps a correlation of 0.081 once the date mean is removed from both the target and `RET_1`. Positive lag-1 autocorrelation in portfolio returns is a known pattern (asynchronous or stale prices, orders executed over several sessions), and it tends to erode once it is traded.

It does erode here. Ordering the main chain in time and cutting it into deciles, the zero-parameter rule falls from 53.3% to 54.7% in the three oldest deciles to between 50.3% and 51.8% in the last six. The lag-1

autocorrelation inside the windows, which can be computed on the test set because it does not use the target, follows the same path, and the test value (0.032) sits in the recent range.



The test set also has 278 allocations per date, like the end of the chain. It looks like the recent regime, not like the average of the history. This is why the protocol scores the recent regime only, and why every local number in this note is expected to overstate the public score by close to one point.

5. Model development

Two ideas passed the decision rule, a third (the blend) was kept on weaker evidence, and everything else failed, including more flexible models. The starting point is a logistic regression on 10 features: yesterday's return and the 5-day and 20-day means, all divided by the window's volatility, plus the volatility itself, yesterday's market move, the day-of-week flag, turnover and group. It scores 51.73% on recent dates, level with the QRT benchmark (51.84%) and its 52 features.

Step	Idea	Recent CV	Gain vs reference at the time	Folds won	Kept
1	Allocation fixed effects, L2-shrunk (C = 0.003)	51.89%	+0.16 pt (t = 5.6)	8/8	yes
2	Cross-sectional features: 5 most correlated peers, group mean, group x group lead-lag, market dispersion	52.21%	+0.32 pt (t = 2.5)	7/8	yes
3	Average with a shallow LightGBM on the same information	52.36%	+0.15 pt (t = 1.7)	7/8	yes, see text
	LightGBM instead of the logistic, same features	52.01%	+0.12 pt vs step 1 (t = 0.8)	4/8	no
	More weight on the recent regime (x2, x4, recent only)	51.83% to 51.90%	-0.06 to 0.00 pt	2 to 4/8	no
	Allocation-specific lag-1 effect	51.88%	none beyond step 1		no
	Residual return vs group, signed-volume pressure	52.32% to 52.34%	-0.02 to -0.04 pt	2 to 3/8	no
	Weighting rows by the size of the move	52.27% to 52.31%	-0.05 to -0.09 pt	1 to 4/8	no
	Nested tuning of the LightGBM	52.31%	-0.05 pt	3/8	no
	Neural network on the raw 20-day window, free / regularised	50.83% / 51.38%	-1.53 / -0.98 pt	0/8	no

Step 1 answers the challenge's question directly. An allocation's share of winning days in the old regime correlates at 0.45 with its share in the recent regime, so some allocations are persistently worth following or fading. Fixed effects with strong shrinkage are the statistical version of an empirical-Bayes estimate of each allocation's base rate. All encodings are fitted inside each fold's training dates.

Step 2 is the main gain. If the sign of each group's mean return for the next day were known, accuracy would reach 62%: a large part of the target is common to a group and a day, and it is read better from neighbours than from the allocation alone. Peer correlations are estimated on each fold's training dates only, otherwise validation targets would leak into the feature through `RET_1` of the following day.

Step 3 does not pass the rule against step 2 alone (t = 1.7, 7 folds out of 8). It was kept because it beats step 1 with t = 2.9 and, on the lockbox, the blend scores 51.98% against 51.66% for the logistic half. The reference for the rejected rows below is the model in force when they were tested: step 1 for the first three, the final blend for the others. The failures are informative too. Model flexibility does not help: the signal is close to linear, and a well-regularised nonlinear model ends up doing what the logistic does, while a neural network fits noise. The nested tuning chose depths from 2 to 4 and 197 to 819 trees depending on the fold, a sign that the configurations differ only by noise.

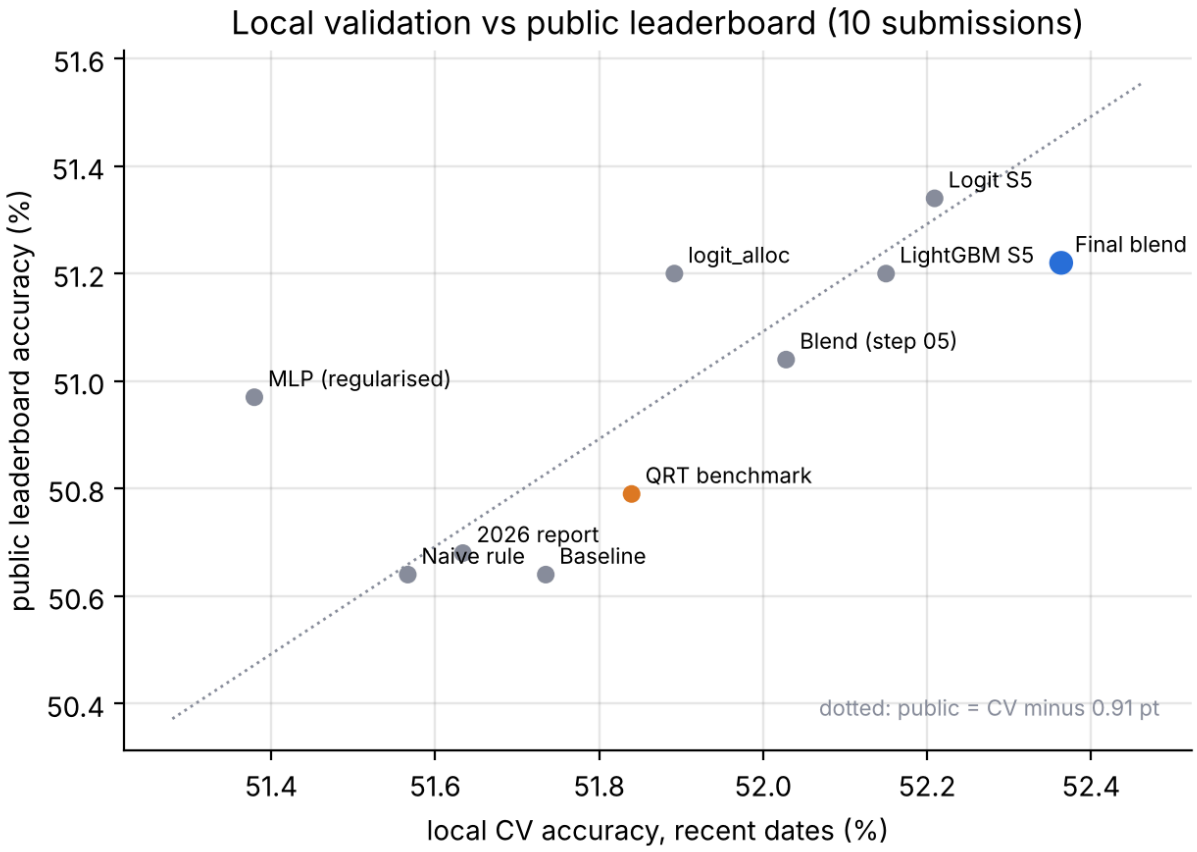
6. Final model and out-of-sample evidence

The final model beats the benchmark on all three measures, and the margin holds on data that played no part in any choice.

Model	**Recent CV**	**Lockbox (250 newest dates)**	**Public leaderboard**
Naive rule sign(`RET_1`)	51.57%	51.10%	50.64%
QRT benchmark (LightGBM, 52 features)	51.84%	51.42%	50.79%
Logistic baseline	51.73%	51.53%	50.64%
Final blend	**52.36%**	**51.98%**	**51.22%**

The gap between CV and lockbox (0.4 point) is the same for the final model as for the benchmark, which suggests that about 50 trials on the same folds did not inflate our scores beyond the known effect of non-stationarity. The lockbox standard error is at least 0.19 point (rows of the same day are correlated), so the 0.56-point margin over the benchmark is encouraging rather than conclusive; its sign agrees with the CV, which was the real test. On the lockbox most of the gain comes from the LightGBM half, while the allocation effects add nothing on that period.

Ten models were submitted to check whether local validation predicts the leaderboard. The rank correlation between recent CV and public score is 0.82 (p = 0.003), with a stable gap of about 0.9 point. The two visible inversions (the logistic half scoring 0.12 point above the blend in public, the regularised network above its CV rank) are well inside the ±0.5 point noise of a public score.



The submission kept for the private leaderboard is the final blend. This was decided before the public scores were known: choosing among close models on about 16,000 noisy public rows would select noise.

7. Comparison with an earlier approach

In March 2026 I worked on the same challenge with a different pipeline. That pipeline was rebuilt from its report and scored with the protocol above; it reproduces the reported figures (52.46% for the model alone and 52.07% for the submitted blend, on all dates, against 52.41% and 52.07% reported).

	Recent CV	**Public**
Earlier pipeline, model alone (LightGBM, ~80 features)	51.95%	not submitted
Earlier pipeline, submitted blend with a momentum vote	51.63%	50.68%
Final blend of this note	52.36%	51.22%

Three lessons came out of the comparison:

- **Block CV on shuffled labels is not temporal.** The earlier pipeline used temporal blocks with a 10-day embargo along the date labels. Since the labels are shuffled, this amounts to a shuffled K-fold.
- **A blend can silence the model.** The submitted formula was 0.65 x model + 0.35 x a momentum vote. With model probabilities between 0.45 and 0.55, the vote decides almost every row: 86.5% of the test predictions equal the vote alone, and the blend scored below the model it was built on.
- **Feature count is not information.** Adding the earlier pipeline's strongest original features (`STREAK`, market breadth, turnover rank) to the final model changed recent CV by -0.02 to +0.02 point. Their information was already present in fewer variables.

8. Limitations

- **The edge is small and decaying.** Even on the lockbox the final model is about 2 points above a coin flip, and the signal it relies on has weakened over the sample. A further decline after the test period would hit every model in this note.
- **The recent regime is defined indirectly.** It is identified by universe size (at least 270 allocations), a proxy for calendar time. If the test period differed in another way, the recent-regime metric would be less representative; the public scores so far support the choice.
- **The order between the four chains is unknown.** Inside a fold, training may use periods later than the validation block, which is why CV still overstates the lockbox by 0.4 point. The lockbox was taken inside the main chain, where the direction of time is certain.
- **The lockbox has been used.** After its single opening, no untouched hold-out remains; later trials were judged on CV and, coarsely, on the public score.
- **Public leaderboard.** Scores near 0.81 on the public leaderboard cannot come from predicting daily return signs. No mechanism reproducing them was found in the data: the test dates have no overlapping windows with any other date. They are left out of every comparison here.

References and reproducibility

- Lo, A. and MacKinlay, A. C. (1990). When are contrarian profits due to stock market overreaction? *Review of Financial Studies*, 3(2). Lead-lag effects and nonsynchronous prices across portfolios.
- López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley. Purged cross-validation and embargo.
- Gu, S., Kelly, B. and Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5).

Code: Python (pandas, NumPy, scikit-learn, LightGBM), fixed seeds. The notebooks run in order: `01_eda`, `02_validation` (chronology and frozen folds, saved to `artifacts/`), `03_baseline`, `04_allocation`, `05_boosting`, `06_final` (final model, lockbox, submission), `07_reseau`, `08_soumissions`, `09_rapport_2026`. Shared functions live in `utils.py`, and the models kept or submitted are logged with their fold scores in `artifacts/journal.csv`. The challenge data is not redistributed.